IOT Data Analytics

Chapter 05

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MHSS Mumbai

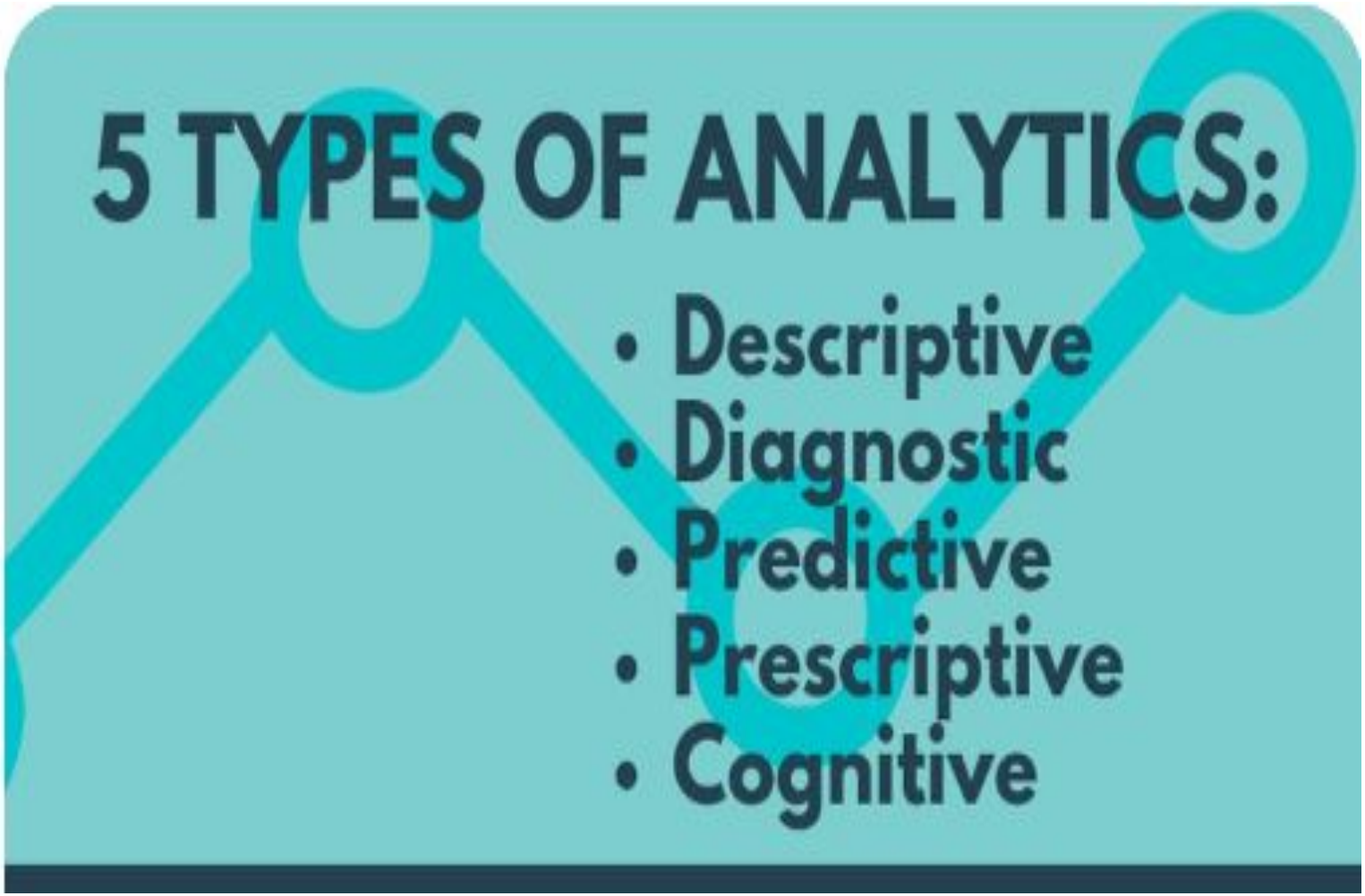
Outline

- Introduction to IoT Data Analytics.
- IoT Analytics Challenges.
- IoT Data Analytics in Cloud Systems.
- Strategies for organizing data for IoT Analytics.
- Linked Analytics Data Set.
- Managing Data Lakes.
- Data Retention Strategies.
- IoT Visualizations & Dashboards.

IoT Analytics

- Internet of Things (IoT) analytics is a data analysis tool that assesses the wide range of data collected from IoT devices.
- IoT analytics assesses vast quantities of data and produces useful information from it.
- IoT analytics are usually discussed in tandem with Industrial IoT (IIoT).
- Data is collected from a wide range of sensors on manufacturing infrastructure, weather stations, smart meters, delivery trucks, and all forms of machinery.
- IoT analytics can be applied to managing data centers and applications handling retail and healthcare.

Analytics Types



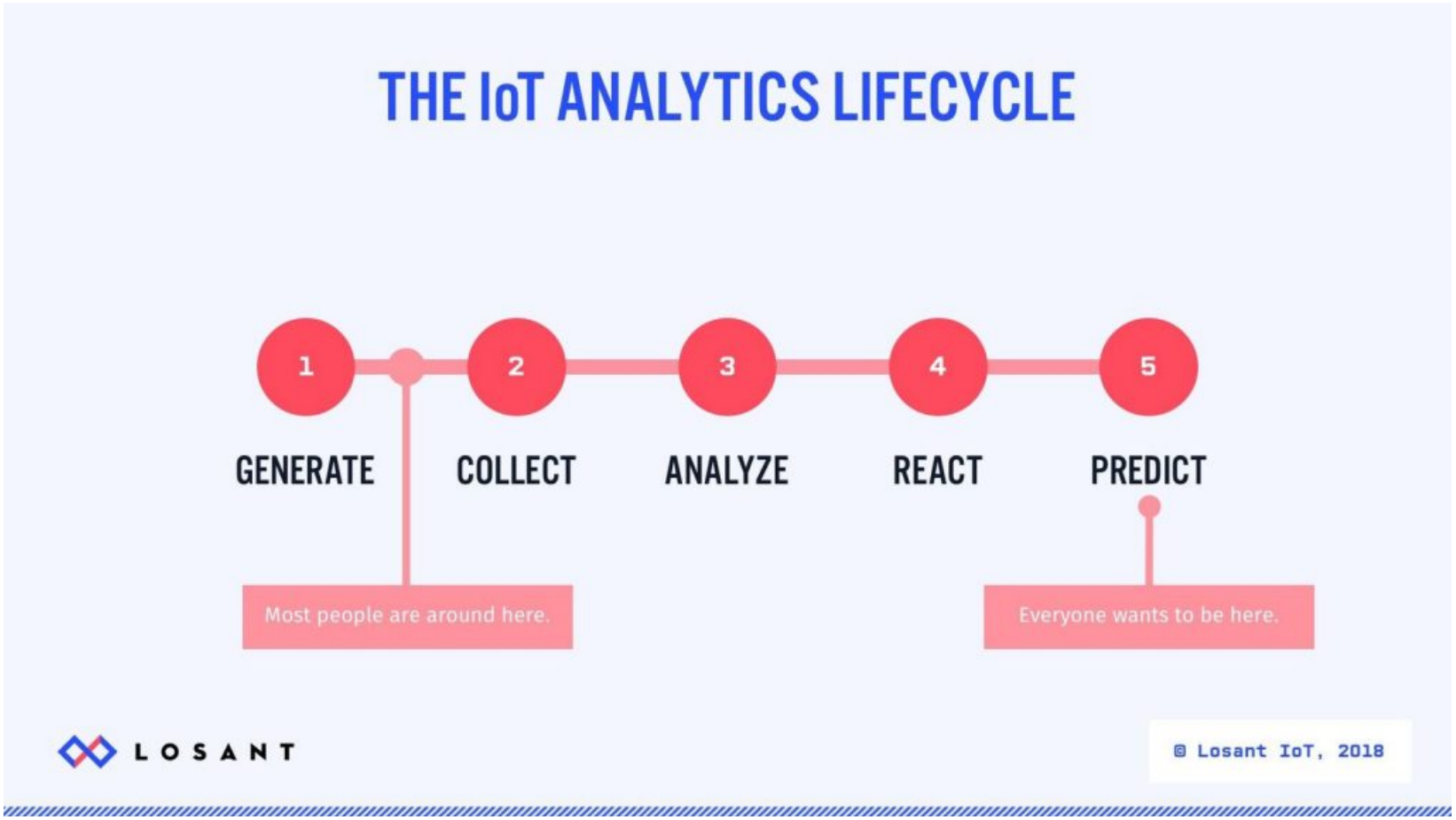
Analytics Types



Analytics Types



IoT Analytics Stages



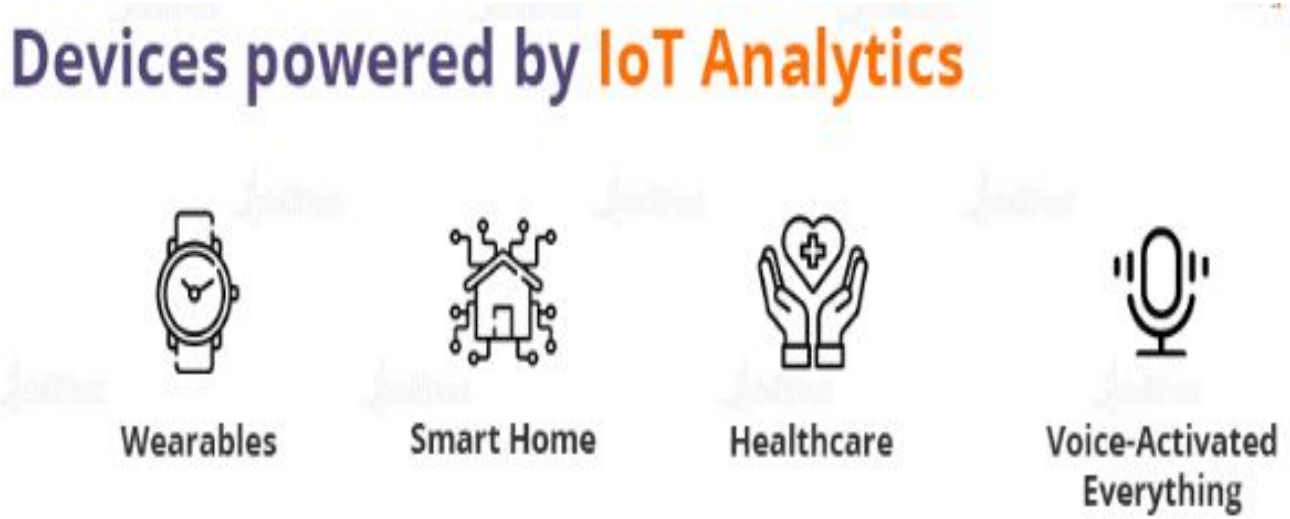
What is IoT Data Analytics

- IoT analytics refers to data analysis tools that analyze the large quantities of data generated by thousands of IoT devices.
- Internet of Things (IoT) analytics is a data analysis tool that assesses the wide range of data collected from IoT devices.
- IoT analytics assesses vast quantities of data and produces useful information from it.
- IoT analytics is the analysis of data generated from connected devices.
- IoT analytics is the application of data analysis tools and procedures to realize value from the huge volumes of data generated by connected Internet of Things devices.

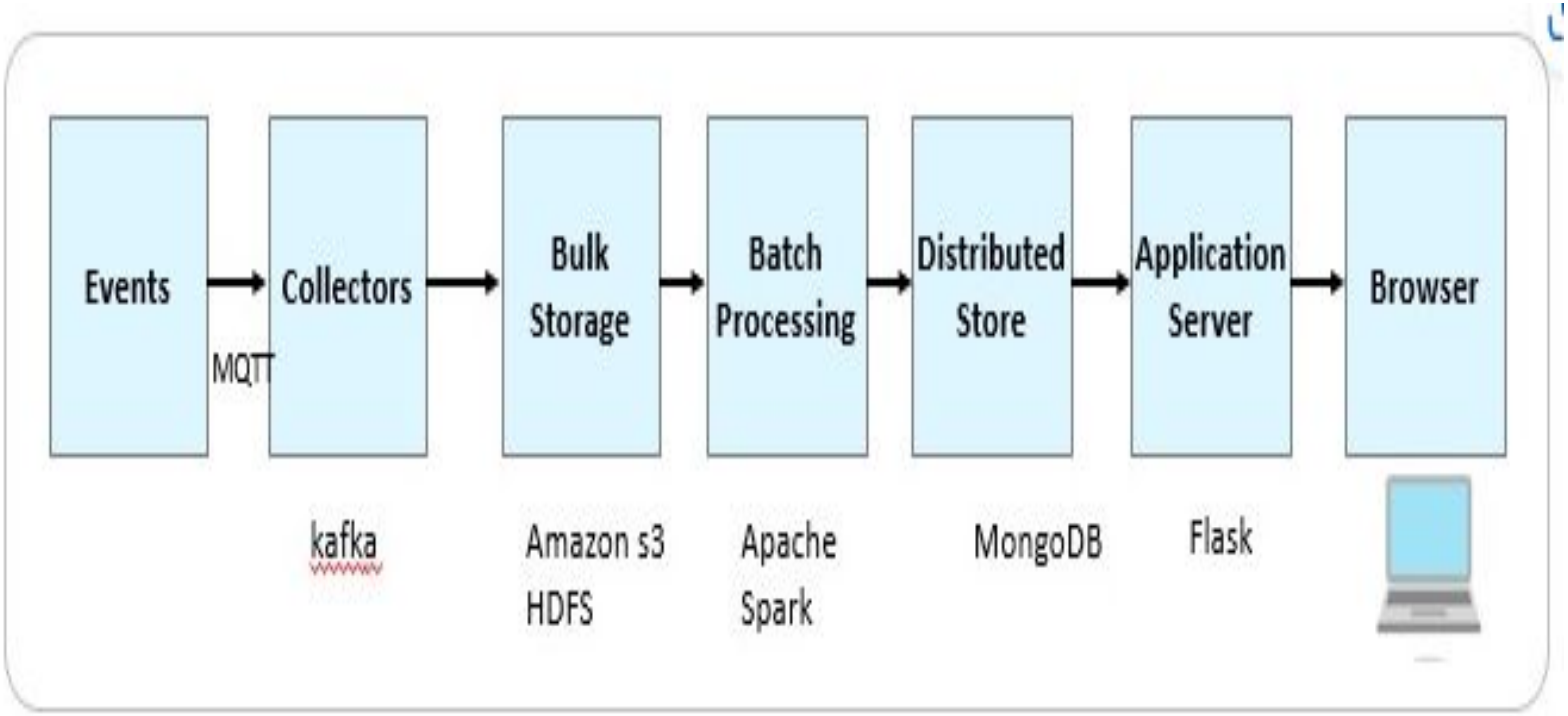
IoT Data Analytics

- IoT analytics are usually discussed in tandem with Industrial IoT (IIoT).
- Data is collected from a wide range of sensors on manufacturing infrastructure, weather stations, smart meters, delivery trucks, and all forms of machinery.
- IoT analytics can be applied to managing data centers and applications handling retail and healthcare.
- In many ways, IoT data is similar to big data.
- The key difference between the two is not just the quantity of data but the range of sources it is obtained from.

IoT Data Sources



IoT Analytics



Benefits of IoT Analytics

- Increased visibility and better control, resulting in quicker decision-making
- Flexible scaling of business needs and expansion in new markets
- Reduction in operational costs from automation and better resource utilization
- New revenue streams, resulting from the clearing of operational challenges
- Quicker solutions from accurate pinpointing of problems
- Quicker resolution of issues, preventing recurrence
- Enhanced customer experience based on analysis of purchase history
- Quicker and more relevant product development.

Challenges in IoT Analytics

- **Excessive data generation and storage requirements** –The aggregate data generated by thousands of IoT sensors are usually very large, thus making it expensive to manage and store such data.
- **The complexity of data** – Data from multiple IoT devices consists of different types, formats and sizes, thus making it very complex and difficult to process and clean.
- **Security** – Businesses, especially those dealing with consumer data, have to take various security measures to protect the stored IoT network data against hacking attempts and leaks.

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Challenges in IoT Analytics

- **Inaccurate data** – Faulty IoT devices lead to inaccurate measurements, thereby messing up the analysis of such data.
- When you have faulty IoT devices at large, the insights offered by IoT analytics tools become unreliable.
- **Building a competent data analysis team** – Businesses need to hire data scientists and analysts to run analytical techniques on the IoT data and derive actionable insights

Challenges BiG Data IoT

- Managing massive amounts of data. It's in the name—big data is big
- Integrating data from multiple sources.
- Ensuring data quality.
- Keeping data secure.
- Selecting the right big data tools.
- Scaling systems and costs efficiently.
- Lack of skilled data professionals.
- Organizational resistance.

IoT Data Analytics

How does IoT analytics work?

IoT analytics work in the following manner:

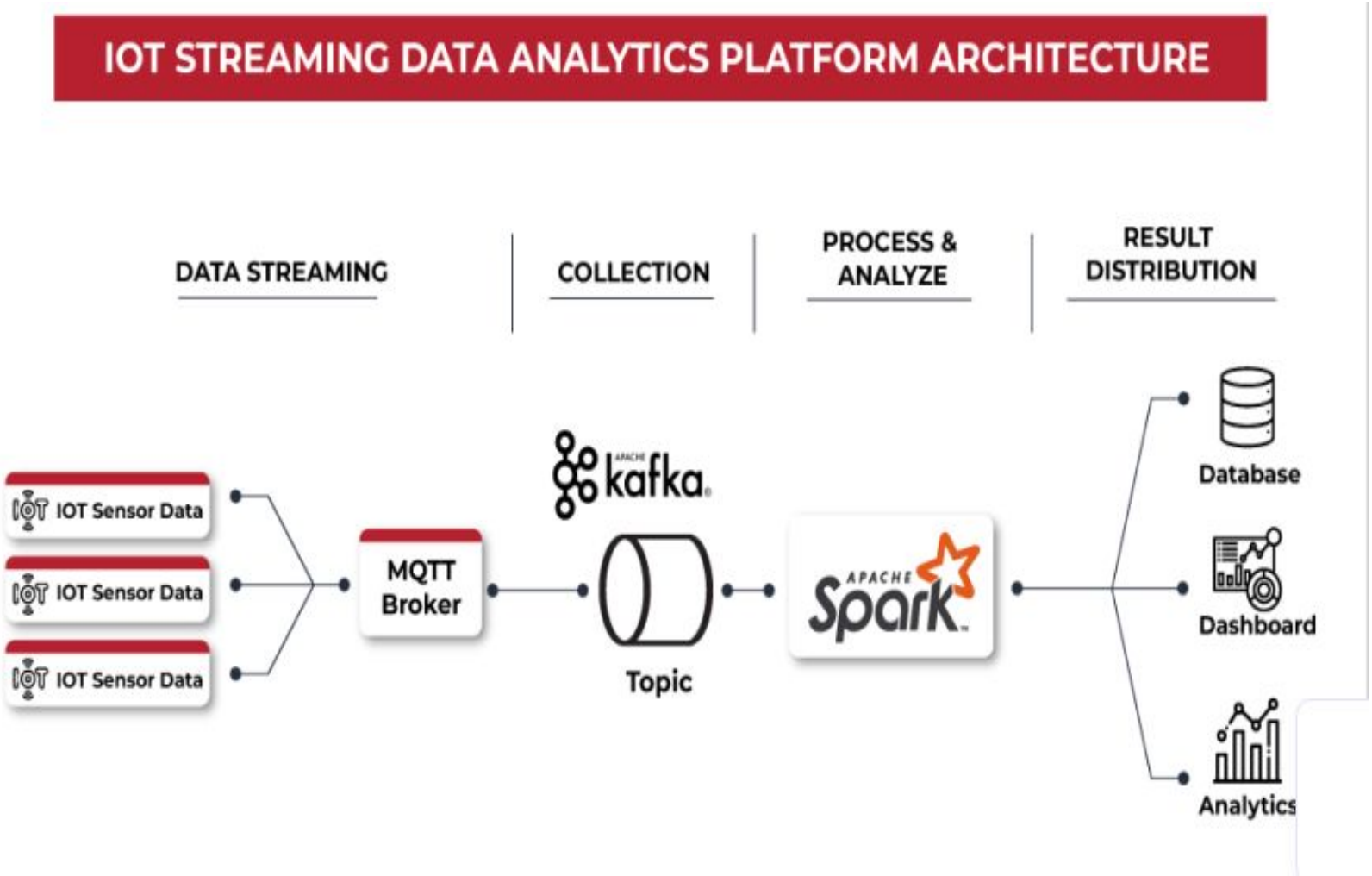
1. Unprocessed raw data recorded by IoT sensors is collected from the entire IoT network. The data is usually in multiple formats.
2. The data collected in step 1 is processed to clean it up for analysis.
3. The cleaned up, processed data is stored in time-series format.
4. IoT analytics tools use various techniques to analyze the time series data and present insights via user-friendly dashboards.
5. Businesses act on these insights to improve their operations.

Types of IoT analytics

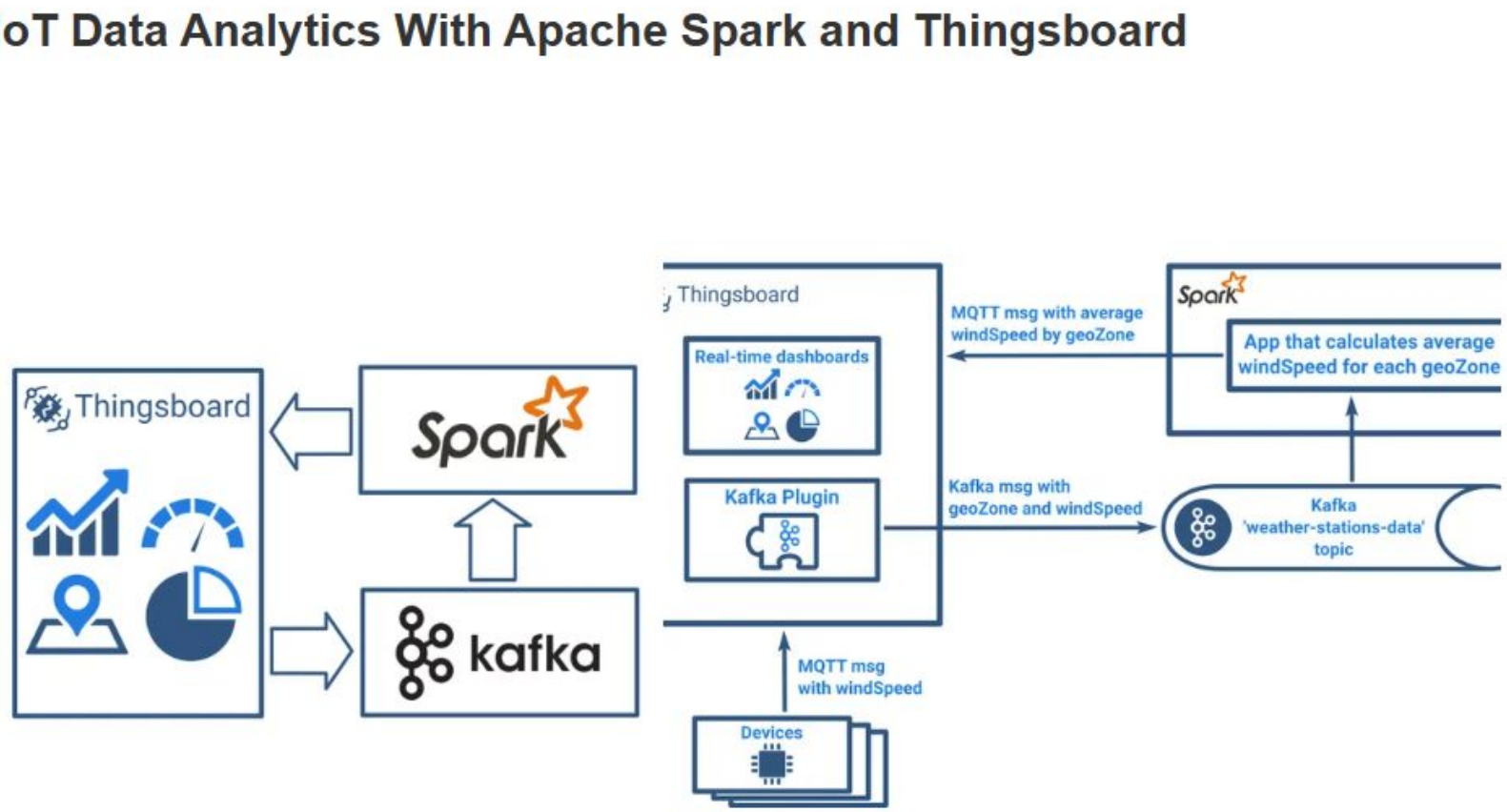
IoT analytics use several big data analytic techniques, such as:

1. **Descriptive analytics**: These analytics indicate whether business operations are going as planned and alert when disruptions occur.
2. **Diagnostic analytics**: These analytics deal with the identification of problems in business operations.
3. **Predictive analytics**: As the name suggests, predictive analytics uses machine learning capabilities and historical data to provide predictions on the performance of business operations and assets.
4. **Prescriptive analytics**: Prescriptive analytics deals with providing actionable steps to take to fix problems in business operations.

IoT Data Analytics



IoT Data Analytics



Strategies of IoT Data Analytics

- A data retention policy clarifies what data should be stored or archived, where that should happen and for how long.
- Once a data set completes its retention period, it can be deleted or moved as historical data to secondary or tertiary storage, depending on business requirements.

Data Vis.

- Data visualization is the representation of data through use of common graphics, such as charts, plots, infographics, and even animations.
- In increasingly data-driven world, it's more important than ever to have accessible ways to view and understand data.
- That's where data visualization comes in handy. With the goal of making data more accessible and understandable.
- Data visualization in the form of dashboards is the go-to tool for many businesses to analyze and share information.

Data Visualization Introduction

1. Introduction

Raw data

Perc.	1963	1983	1989	1992	1995	1998	2001	2004	2007	2010	2013	2016
1	-6405	-7939	-15151	-18951	-19967	-33773	-29129	-32787	-36200	-88315	-81354	-80620
2	-4430	-3542	-8975	-10985	-11322	-20804	-16041	-17423	-21771	-56902	-54339	-50920
3	-3458	-1932	-6437	-5954	-7360	-12617	-9010	-11065	-12437	-37913	-38615	-34580
4	-2689	-727	-3526	-3807	-4213	-8440	-4918	-6969	-8222	-26959	-25778	-25749
5	-2080	-85	-1586	-1895	-2349	-4426	-2100	-3459	-5327	-17022	-19127	-17300
6	-1514	0	-933	-839	-940	-2375	-677	-1488	-2281	-11882	-13465	-11710
7	-1003	0	-224	-84	-16	-885	0	-216	-892	-7936	-9331	-7998
8	-548	0	0	0	0	0	0	0	0	-4974	-6290	-5170
9	-295	302	0	0	0	0	0	1	0	-2885	-3908	-2818
10	-18	724	0	0	94	0	135	254	35	-1105	-2114	-950
11	6	1219	0	168	501	148	542	1145	359	-66	-691	0
12	6	1661	93	553	1175	472	1192	1908	1100	0	0	60
13	6	1992	373	1140	1724	885	2032	2416	1853	0	1	320
14	6	2415	746	1509	2553	1608	2845	2951	2571	133	206	680
15	160	2873	1120	2012	3461	2287	3658	3561	3474	774	557	1100
16	437	3460	1866	2683	4353	3143	4471	4197	4354	1470	1237	1560
17	794	4023	2146	3254	5481	4131	5419	5062	5362	2211	2011	2350
18	1144	4754	2799	3857	6421	5238	6612	5977	6369	3207	2939	3230
19	1298	5469	3732	4746	7172	6123	7859	6931	7353	3869	3712	4080
20	1821	6293	4105	5920	8143	7259	9213	8228	8488	4753	4434	4800
21	2314	7029	4777	6960	9286	8263	11042	9780	9496	5598	5186	5620
22	2793	7855	5803	8221	10805	9369	12194	11319	11349	6356	6094	6410
23	3156	9176	7277	9516	12215	11066	13725	13036	13028	7140	7094	7651
24	3839	10480	8415	11069	13483	12763	15554	14880	14568	8124	8259	8971



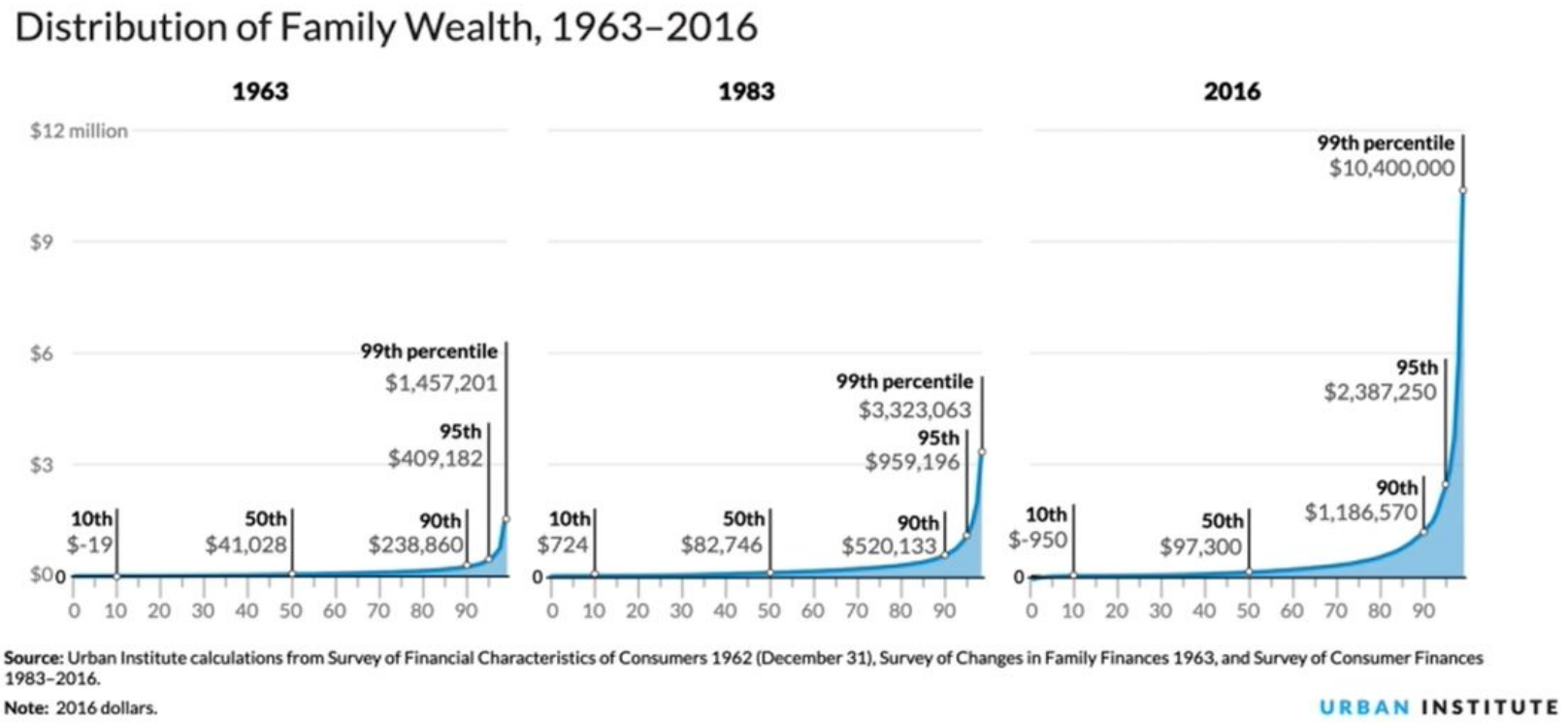
Data



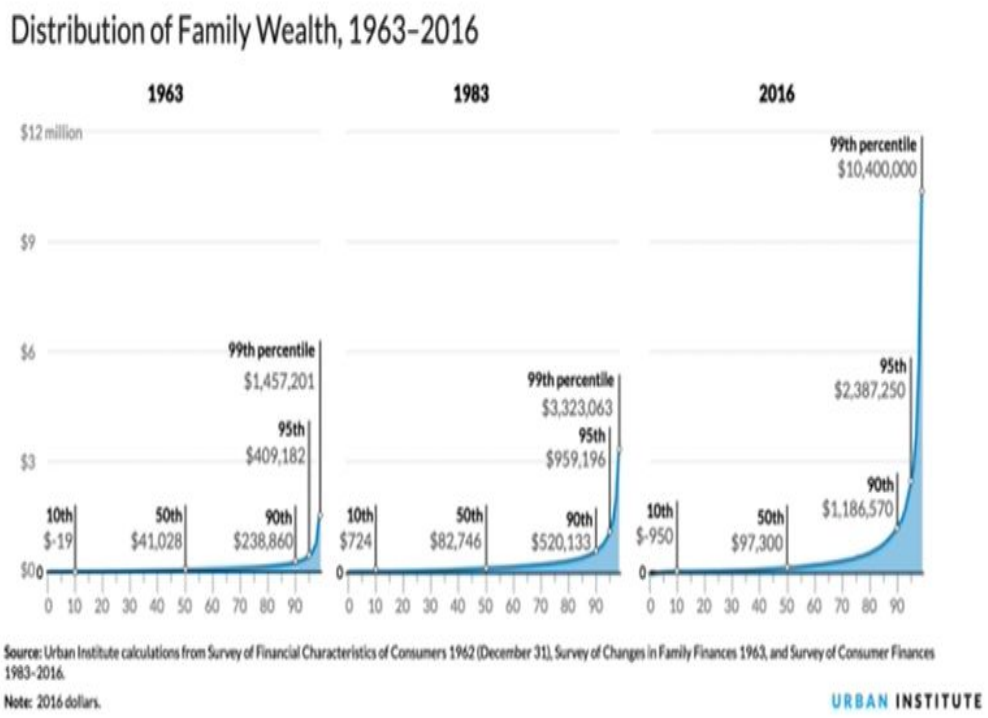
Message

Communication fails

Visual Description on Data



Visual Description of Data



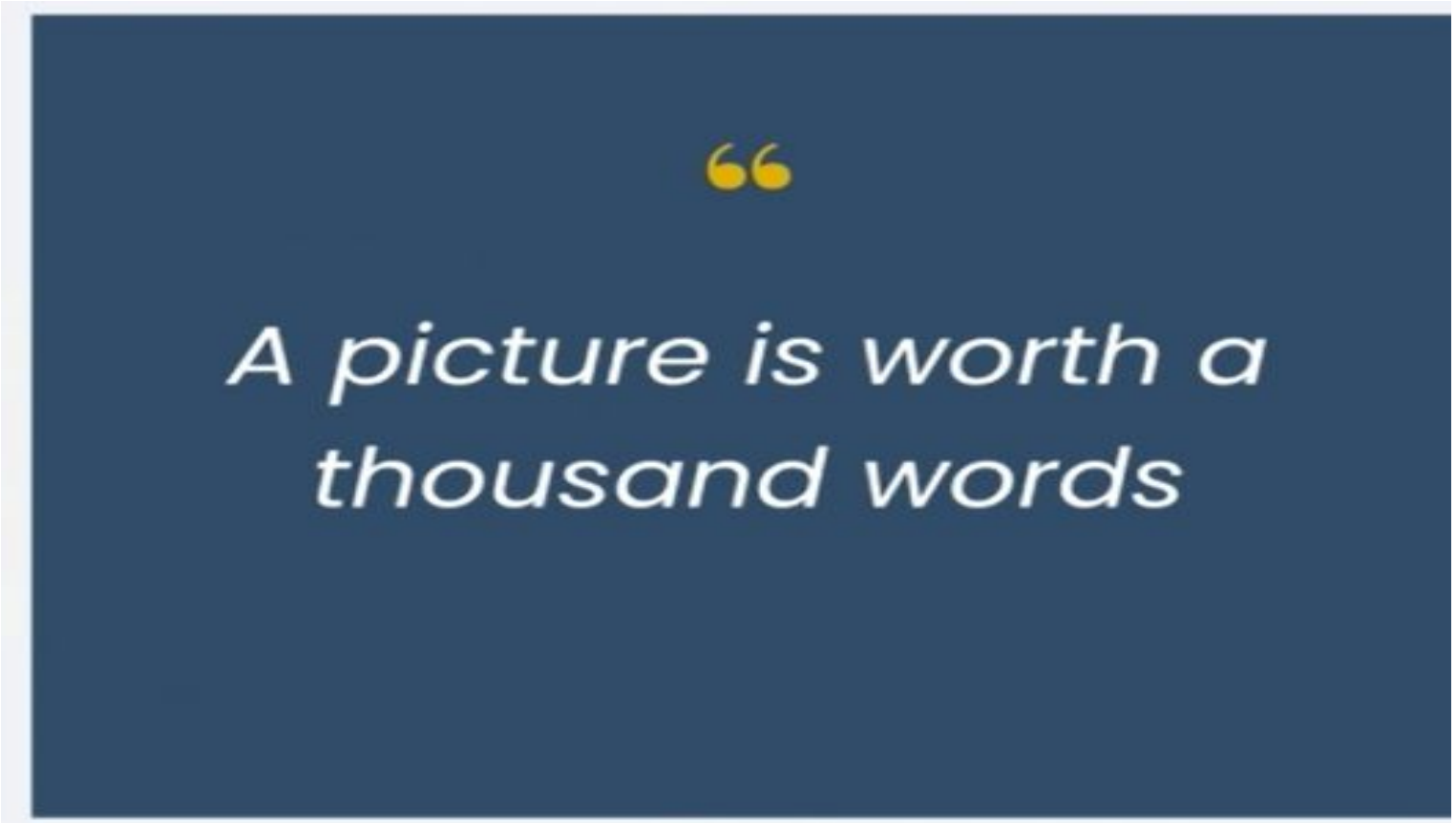


Message

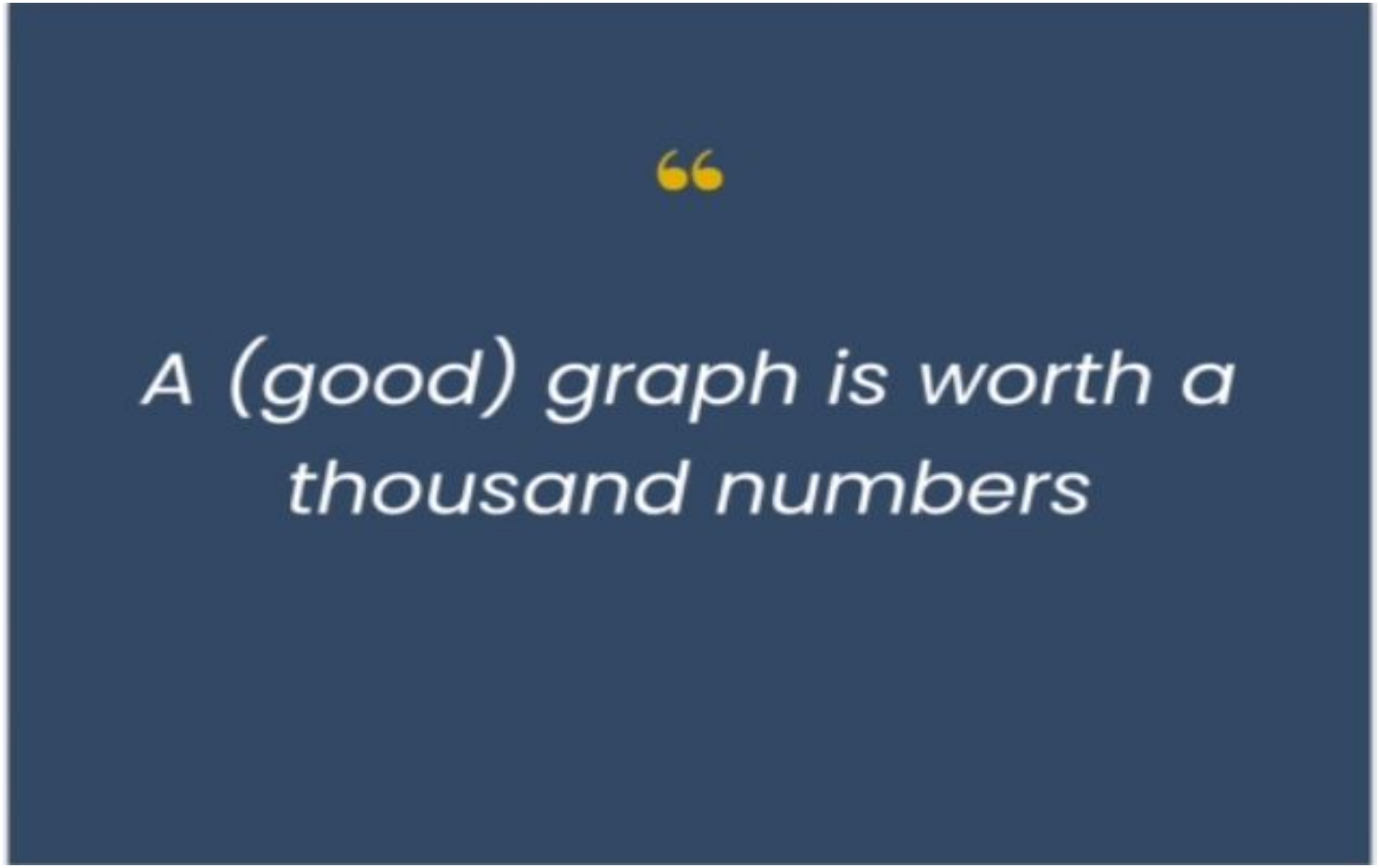


Data

Data Visualizations Introduction



Data Visualization Introduction



Why Data Visualization

Communicate data in a natural, effective way

Reveal information in a more impactful way

Assist the viewer in **thinking** about the content

why

- Data visualization provides a quick and effective way to communicate information in a universal manner using visual information.
- The practice can also help businesses identify which factors affect customer behavior; pinpoint areas that need to be improved or need more attention;
- The ability to absorb information quickly, improve insights and make faster decisions.
- An easy distribution of information that increases the opportunity to share insights with everyone involved.
- Eliminate the need for data scientists since data is more accessible and understandable; and
- An increased ability to act on findings quickly and, therefore, achieve success with greater speed and less mistakes.

Data Vis


Entire View
Show Me
Data
Analytics
Summarize
Constant Line
Average Line
Median with Quartiles
Box Plot
Totals
Model
Average with 95% CI
Median with 95% CI
Trend Line
Forecast
Cluster
Custom
Reference Line
Reference Band
Distribution Band
Box Plot
Marks
Circle
Color
Size
Label
Detail
Tooltip
Columns
Shoulder Breadth
Rows
Bust Chest Circumfer..
Bust Chest Circumference
140
130
120
110
100
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Shoulder Breadth

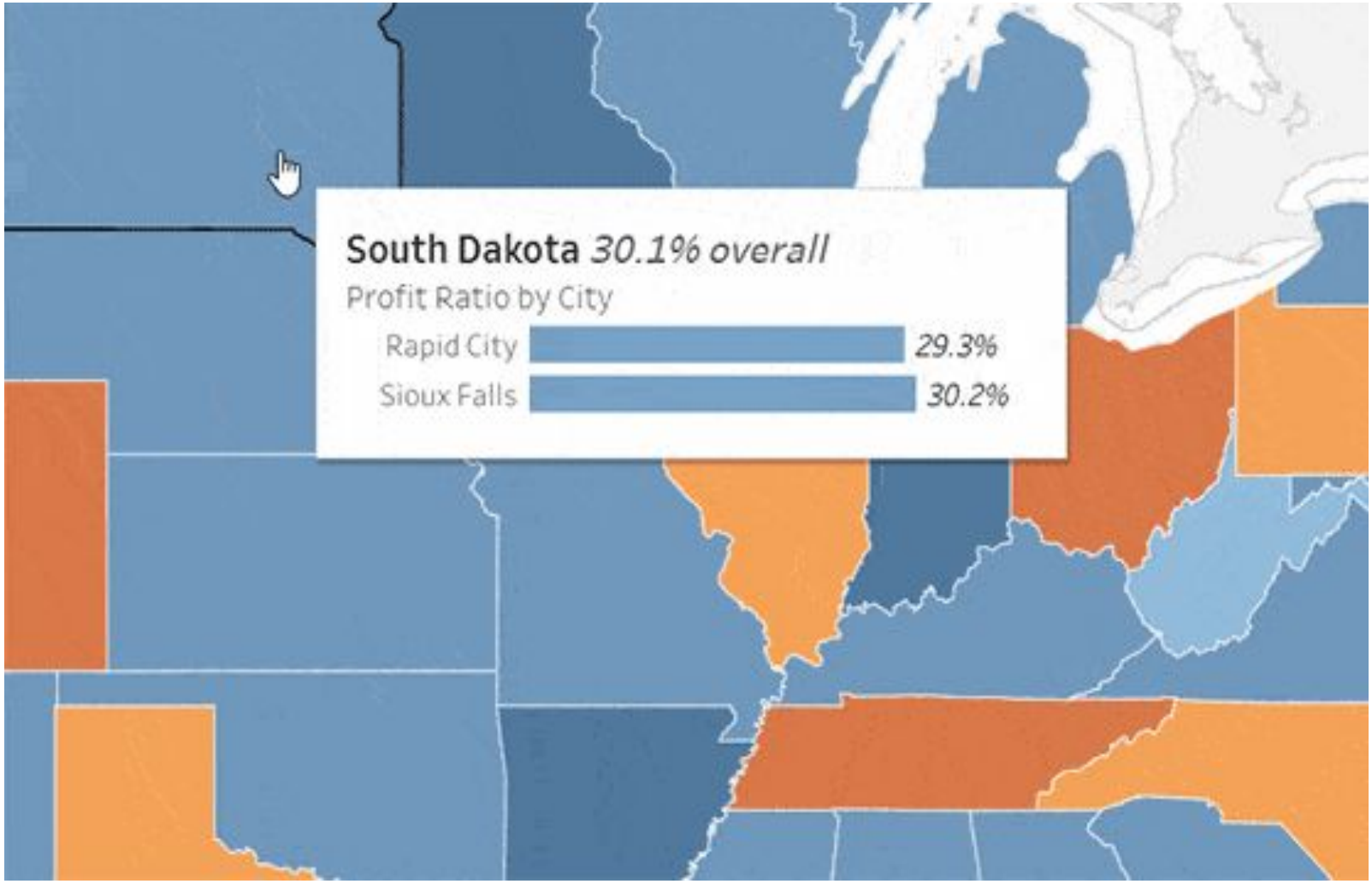
Types

- **Chart:** Information presented in a tabular, graphical form with data displayed along two axes. Can be in the form of a graph, diagram, or map.
- **Table:** A set of figures displayed in rows and columns.
- **Graph:** A diagram of points, lines, segments, curves, or areas that represents certain variables in comparison to each other, usually along two axes at a right angle.
- **Geospatial:** A visualization that shows data in map form using different shapes and colors to show the relationship between pieces of data and specific locations.

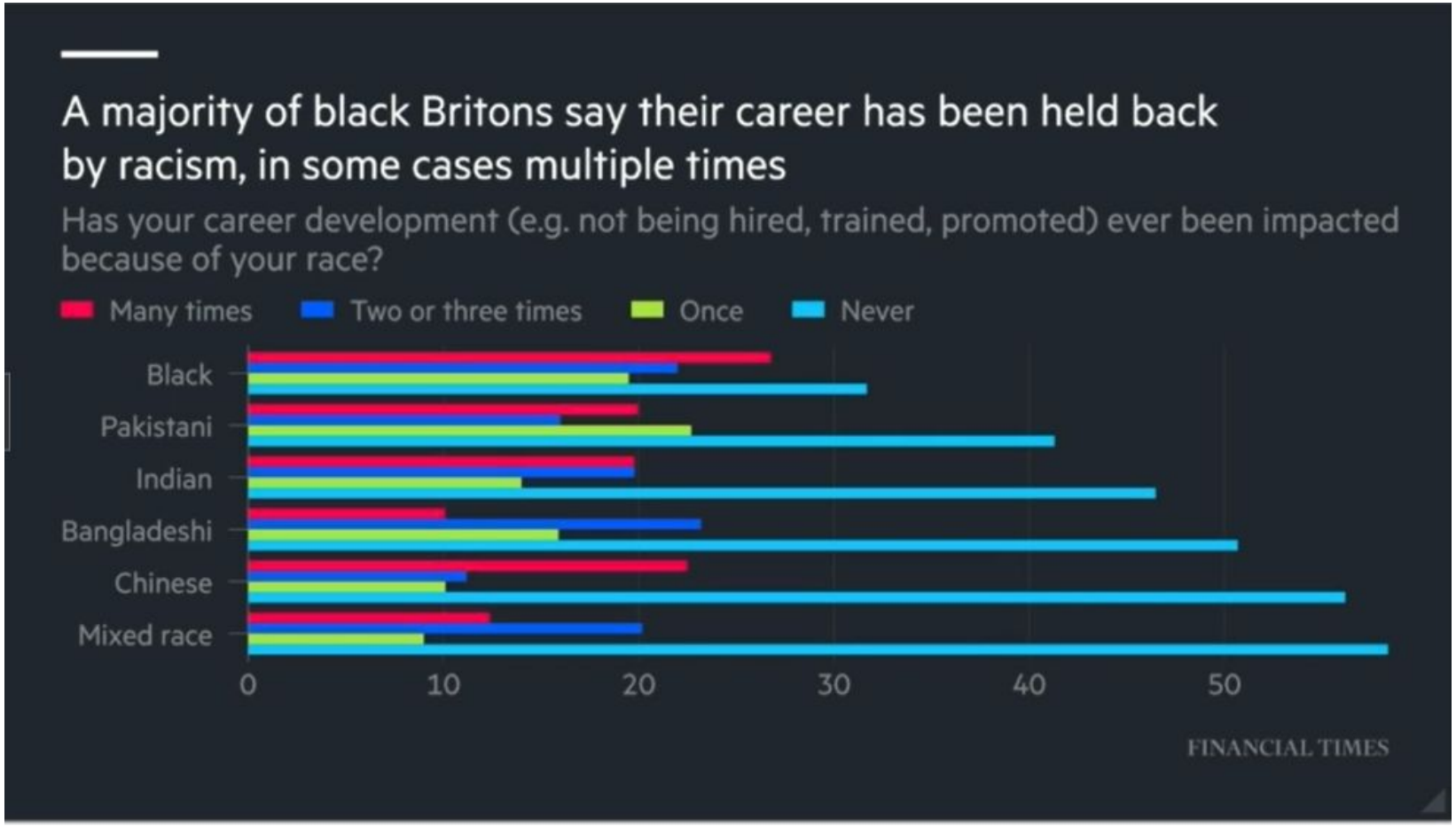
Types

- **Infographic:** A combination of visuals and words that represent data. Usually uses charts or diagrams.
- **Dashboards:** A collection of visualizations and data displayed in one place to help with analyzing and presenting data.

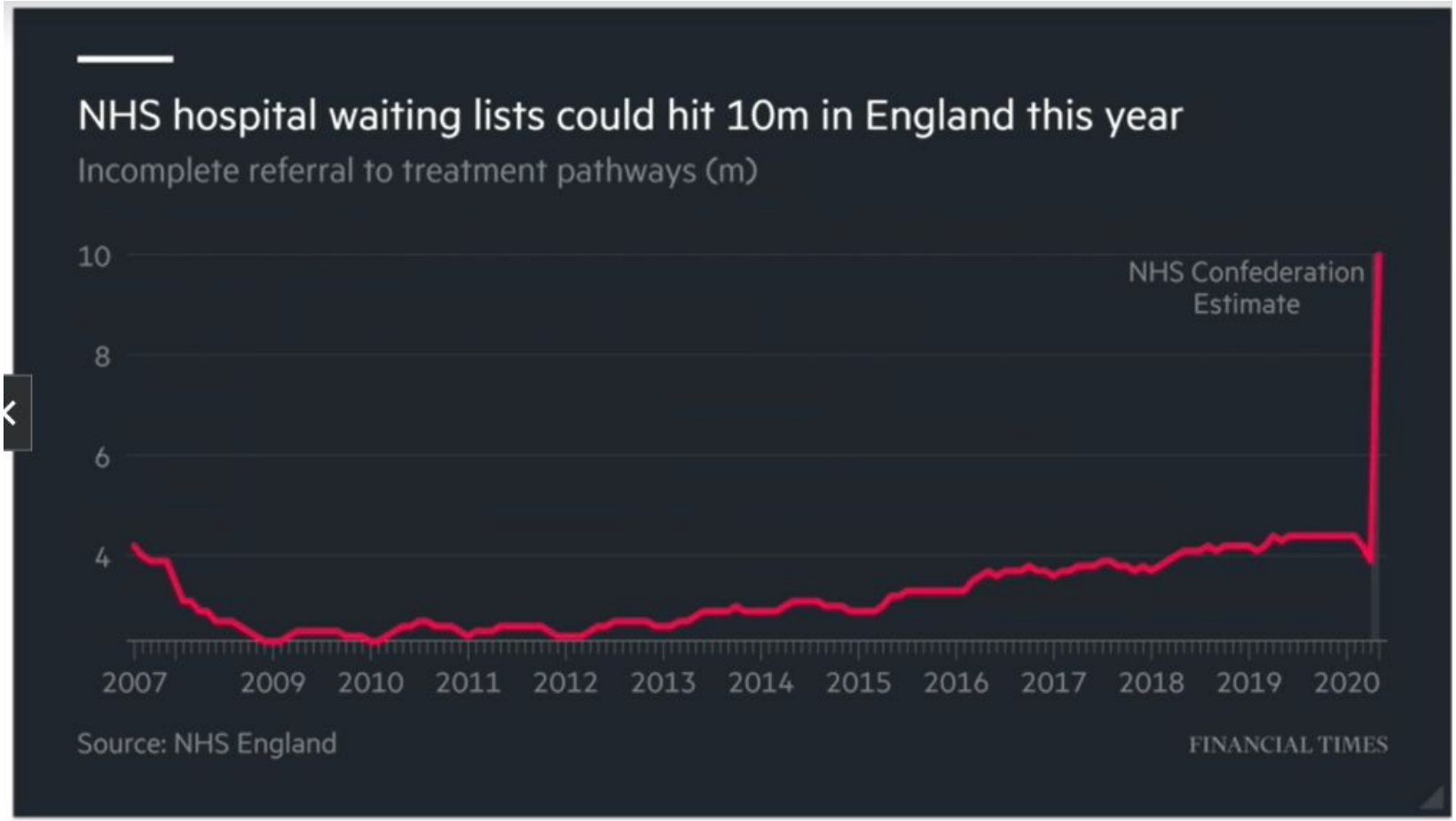
Vis.


South Dakota 30.1% overall
Profit Ratio by City
Rapid City
29.3%
Sioux Falls
30.2%

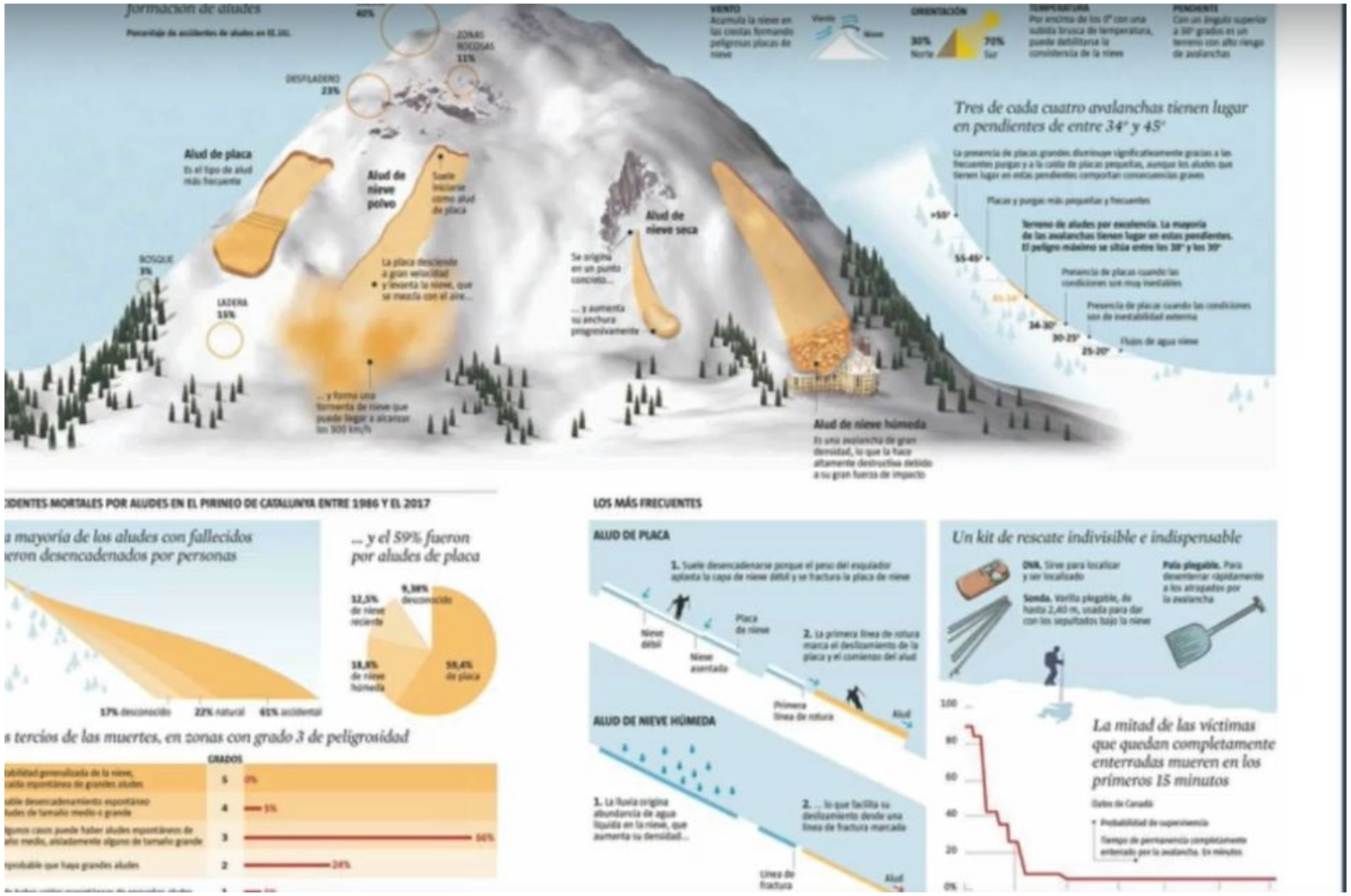
Examples News paper



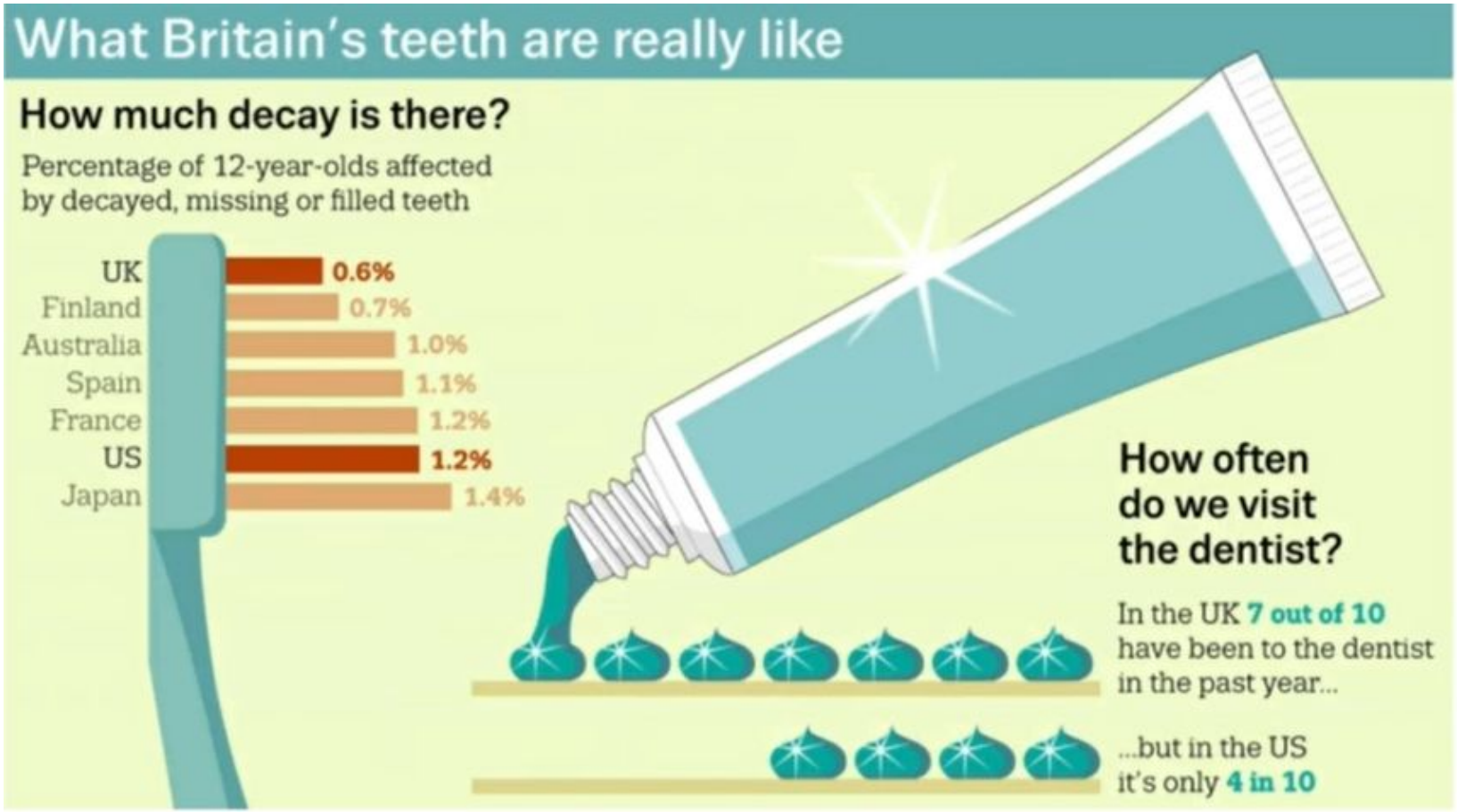
Example Newspaper



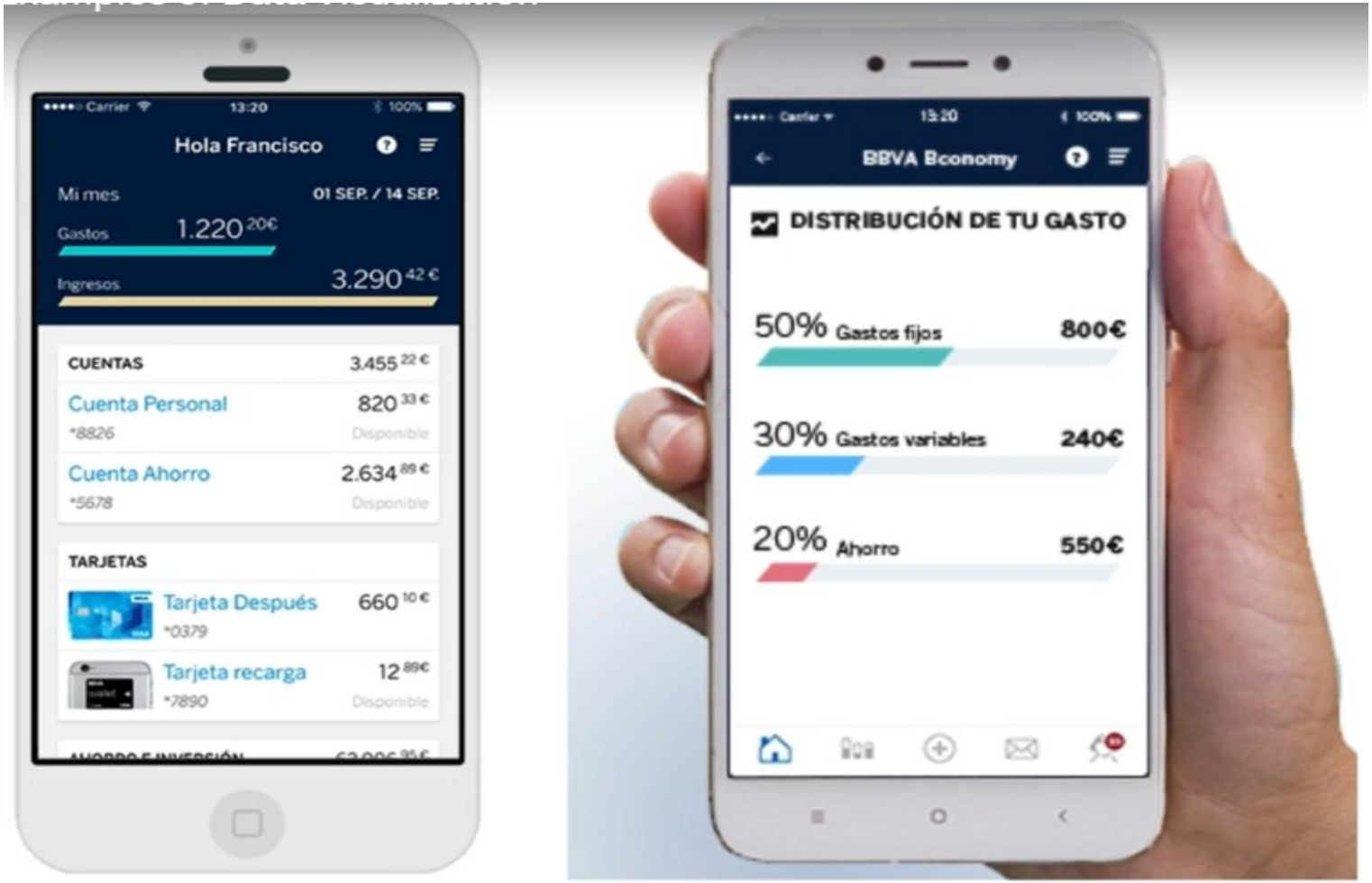
Example Info Graphic



Example Info Graphic



Example UI/ UX Design



Example UI/UX Design



ADIDAS RUNTASTIC APP

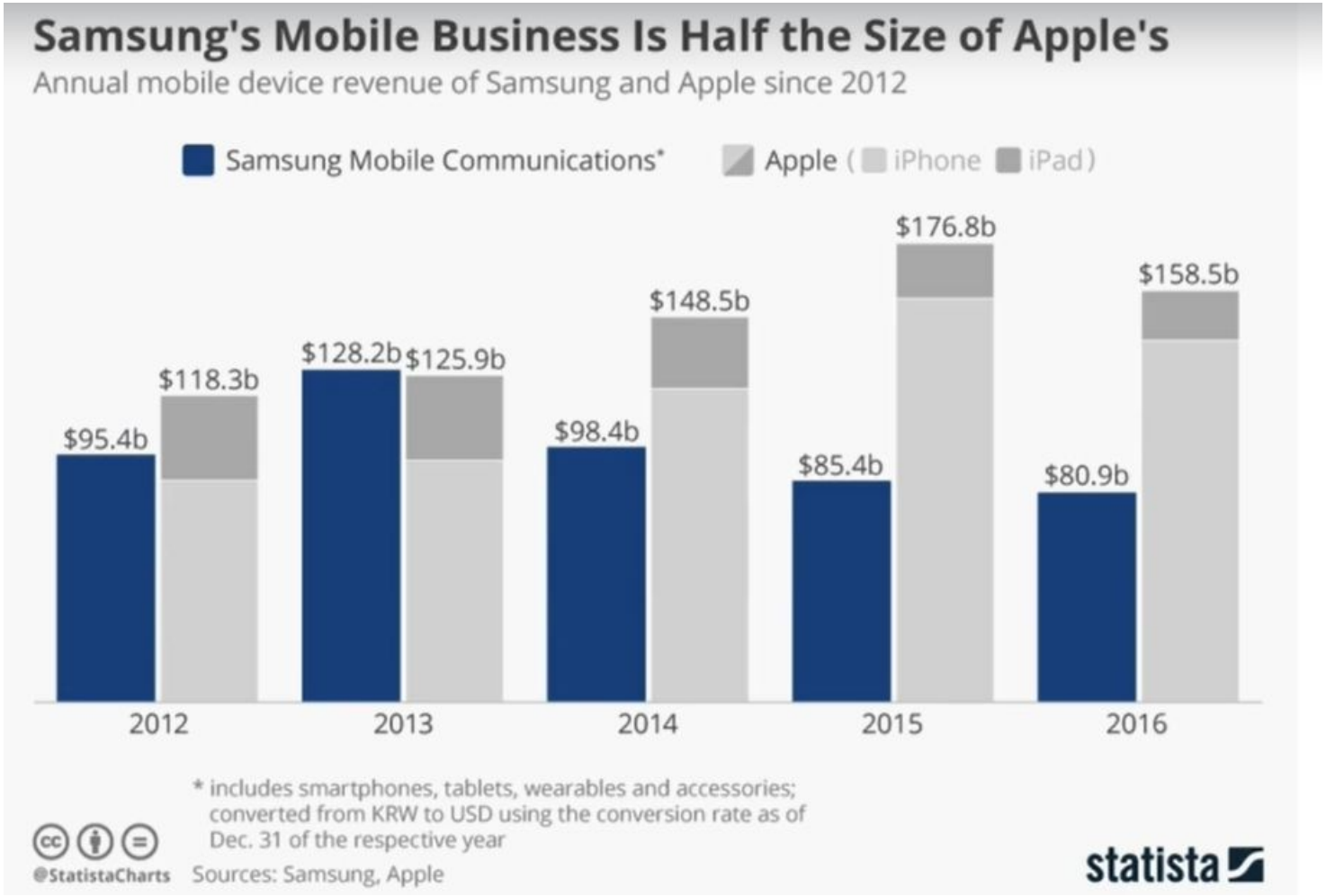


IWATCH

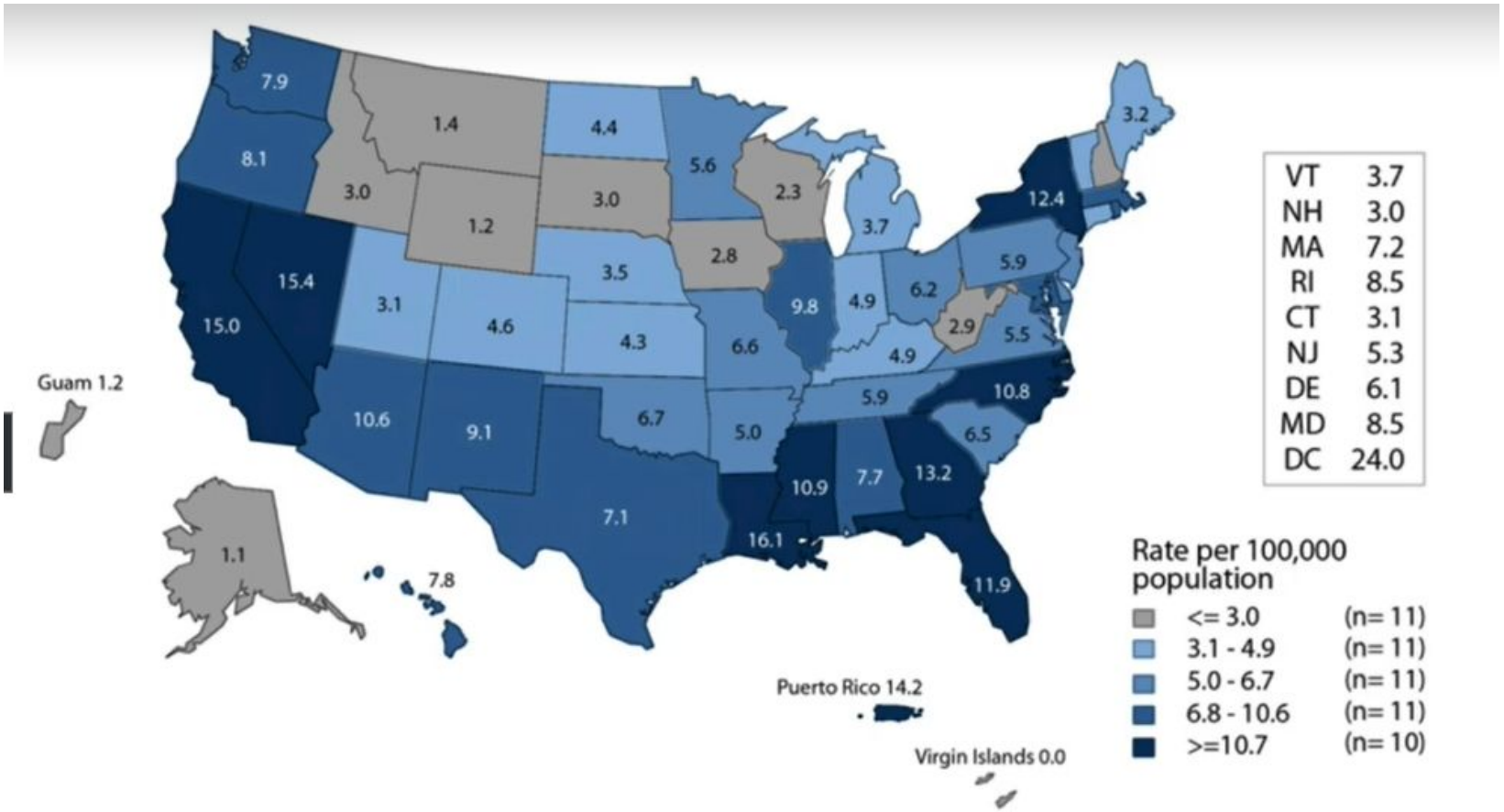
Website Visualization



Business Reports



Health Report



Scientific Papers

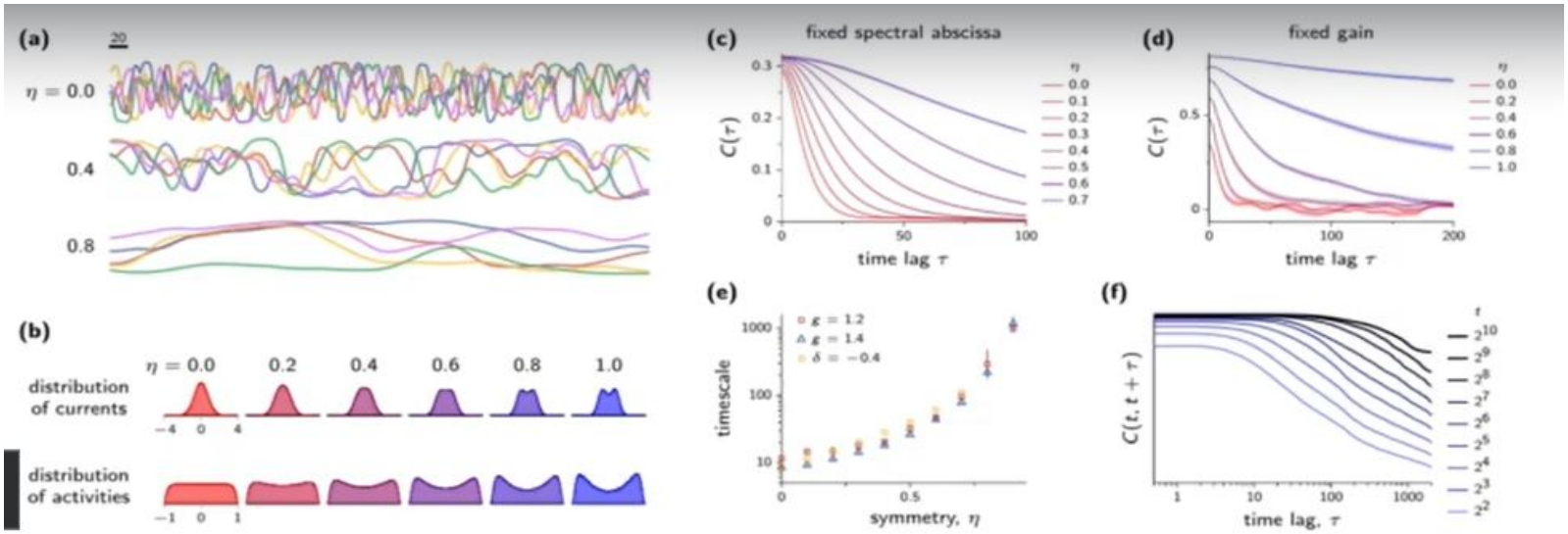
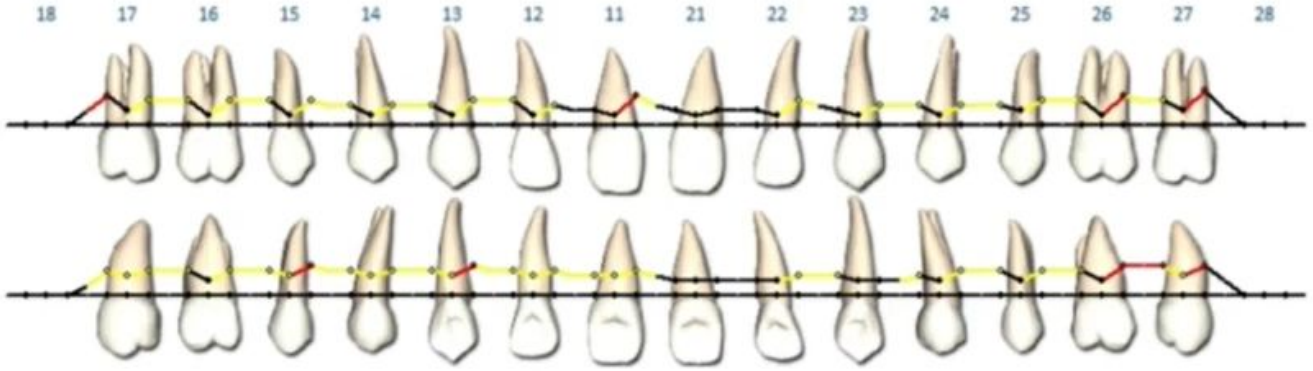


FIG. 5. Effects of symmetry in the chaotic regime. (a) Firing rates of five arbitrary units in a network of size $N = 10^4$, for three different values of η (value indicated next to each inset). The spectral gap is $\delta = -0.4$ (i.e., the spectral abscissa is 0.4). (b) Histograms of the currents x and firing activities $\phi(x)$, for different values of η. Samples were taken every 40 time units from the simulated activity of a network of 10^4 units for ten different realizations of the connectivity matrix and $\delta = -0.4$. (c) Population-average autocorrelation, for different values of the symmetry parameter η and $\delta = -0.4$. The autocorrelation was estimated from the simulated activity of a network of 10^4 units, using 200 different draws of the connectivity matrix. The estimated standard error of the mean is shown in semitransparent shade. (d) Same as in (c), but keeping the gain fixed at $g = 1.5$ instead of fixing δ. For this panel we estimated the autocorrelation using ten independent realizations of the connectivity matrix. Notice the different axis ranges with respect to (c). (e) Timescale of the network fluctuations, estimated from the width of the autocorrelation function at half of its maximum value, using the simulation results from (c) and (d). The error bars indicate the standard error of the mean. (f) Nonstationarity of the network activity for $\eta = 1$. The autocorrelation function depends on two timescales, the time lag τ and the time t, since the simulation started with an arbitrary initial condition. The autocorrelation is estimated from simulated neuronal activity of size $N = 10^4$ for $T = 3030$ time units, using five different trials with different realizations of the connectivity matrix. In this figure the gain was fixed at $g = 1.4$.

Dentist

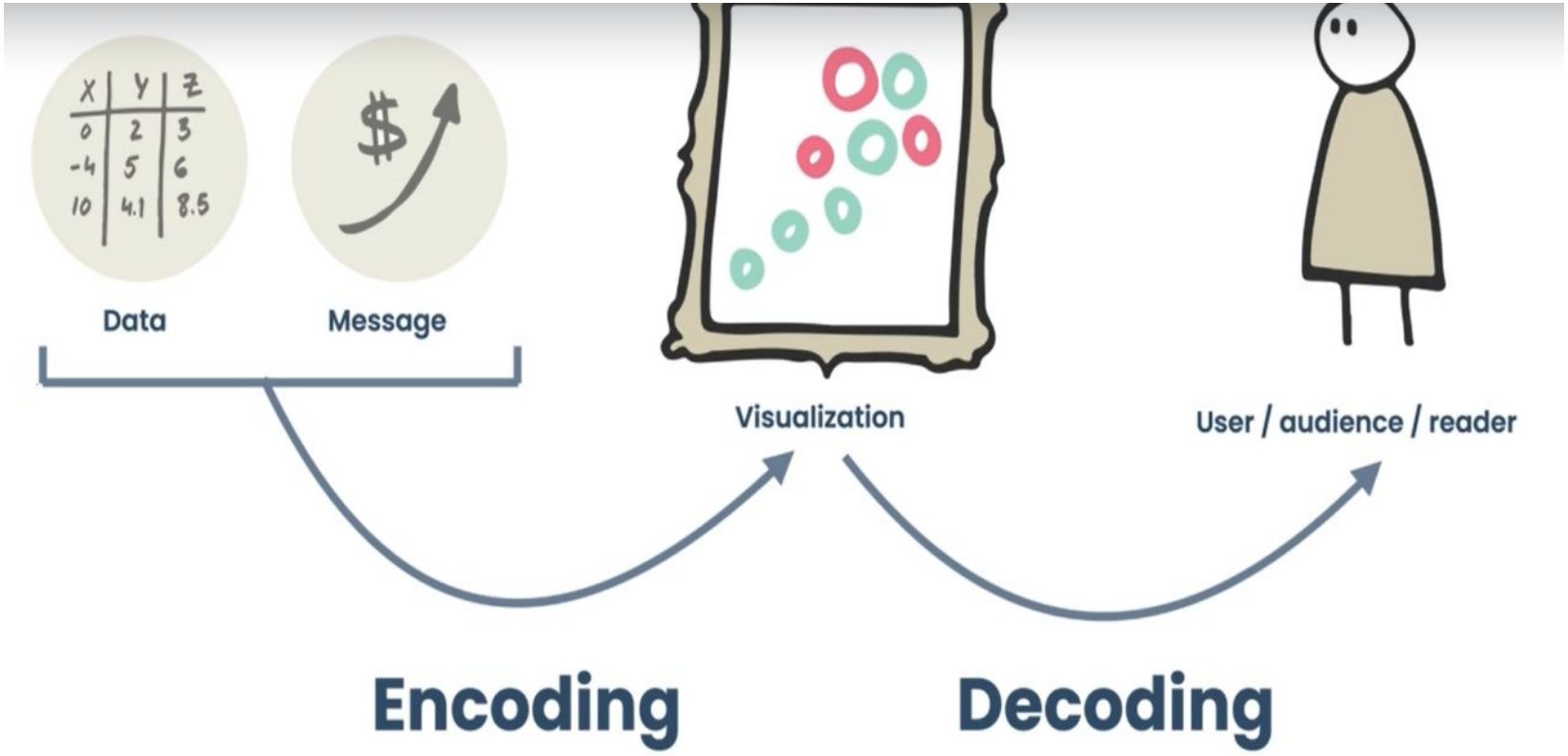
	18	17	16	15	14	13	12	11	21	22	23	24	25	26	27	28
Profundidad bolsa vestibular	0, 0, 0	6, 3, 5	5, 2, 5	5, 2, 5	4, 2, 4	4, 2, 5	5, 2, 4	3, 2, 6	3, 2, 3	3, 2, 5	3, 2, 4	4, 2, 4	4, 3, 5	5, 2, 6	5, 3, 7	0, 0, 0
Recesión vestibular	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0
Pérdida inserción vestibular	0, 0, 0	6, 3, 5	5, 2, 5	5, 2, 5	4, 2, 4	4, 2, 5	5, 2, 4	3, 2, 6	3, 2, 3	3, 2, 5	3, 2, 4	4, 2, 4	4, 3, 5	5, 2, 6	5, 3, 7	0, 0, 0
Linea mucogingival	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Índice de sangrado 0%																



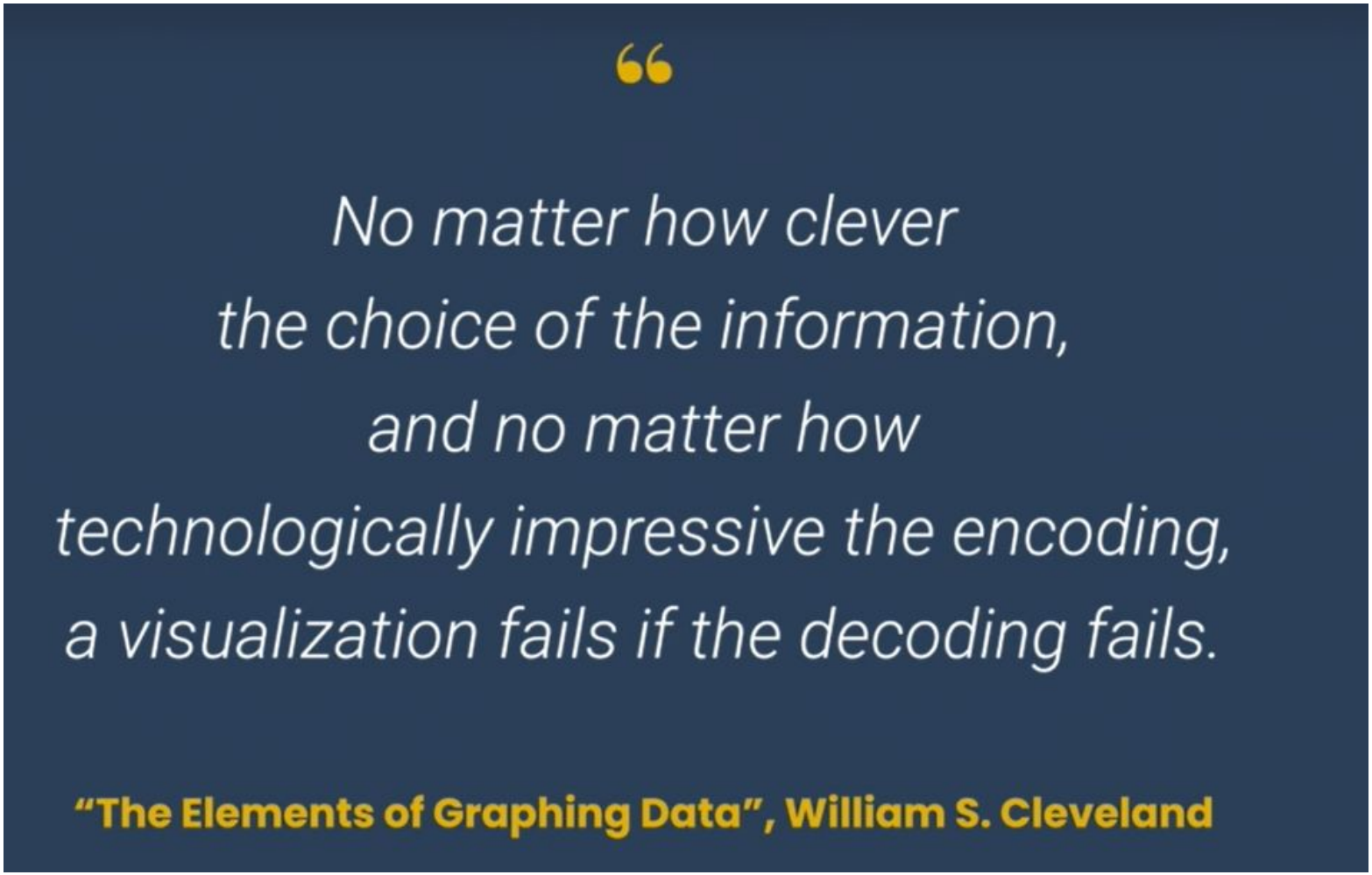
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	18	17	16	15	14	13	12	11	21	22	23	24	25	26	27	28
Profundidad bolsa palatina	0, 0, 0	5, 4, 5	5, 3, 5	5, 4, 6	5, 4, 5	5, 4, 6	5, 4, 5	4, 4, 5	3, 3, 3	3, 3, 4	4, 3, 3	4, 3, 5	5, 3, 5	5, 3, 6	6, 4, 6	0, 0, 0
Recesión palatina	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0	0, 0, 0
Pérdida inserción palatina	0, 0, 0	5, 4, 5	5, 3, 5	5, 4, 6	5, 4, 5	5, 4, 6	5, 4, 5	4, 4, 5	3, 3, 3	3, 3, 4	4, 3, 3	4, 3, 5	5, 3, 5	5, 3, 6	6, 4, 6	0, 0, 0

Data Vis.


X Y Z
0 2 3
-4 5 6
10 4.1 8.5
Data
Message
Visualization
User / audience / reader
Encoding
Decoding

Data Vis.


“
No matter how clever
the choice of the information,
and no matter how
technologically impressive the encoding,
a visualization fails if the decoding fails.
“The Elements of Graphing Data”, William S. Cleveland

Which One Better?

SHOW THE DATA

- Remember that the goal of a visual display is to inform about some data.
- Do not hide it, do not overprocess it.





Which One Better?

Induce the viewer to think about the substance rather than about methodology, graphic design, the technology of graphic production, or something else

- Do not conceal the data with elements that steal the attention away from the information.

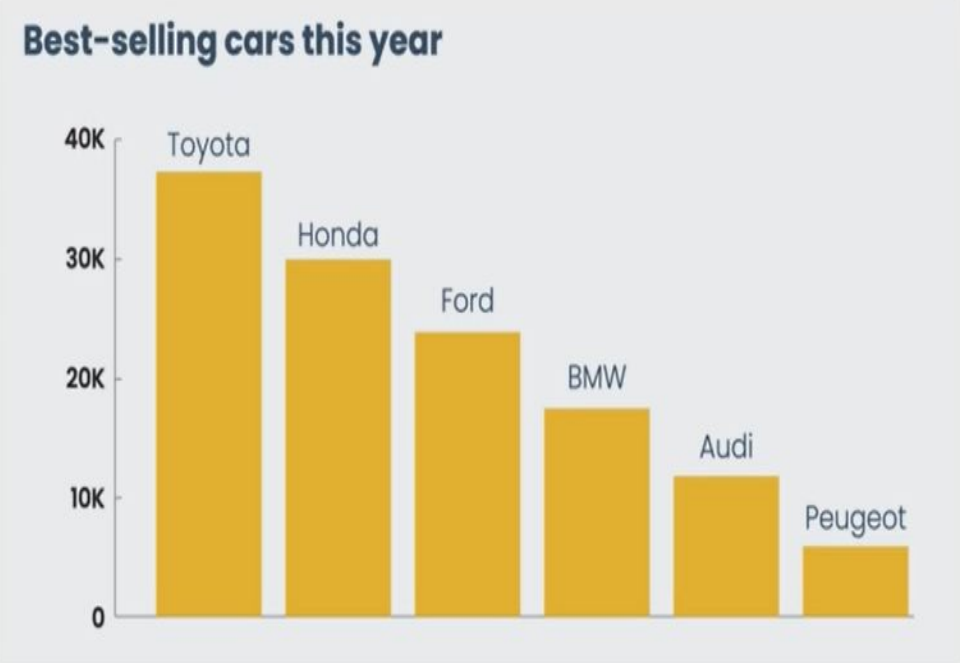


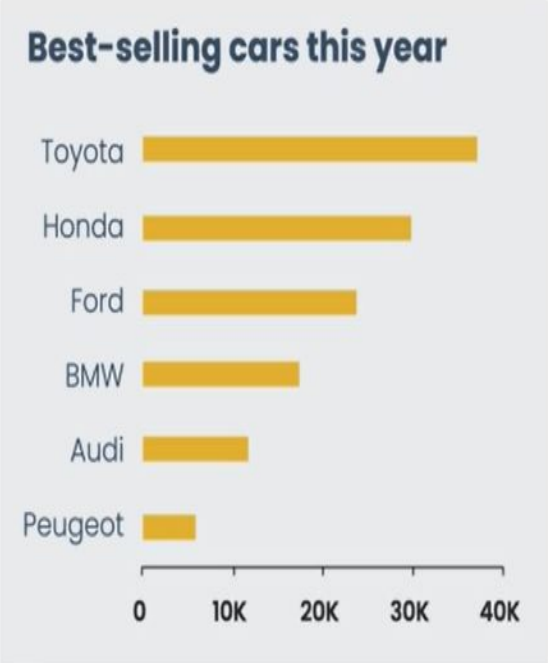


Which One is better

Present many data in a small space

- Do not waste a very big space for very few data.
- Space might be scarce. Too much space gives the reader no clear focus of attention.

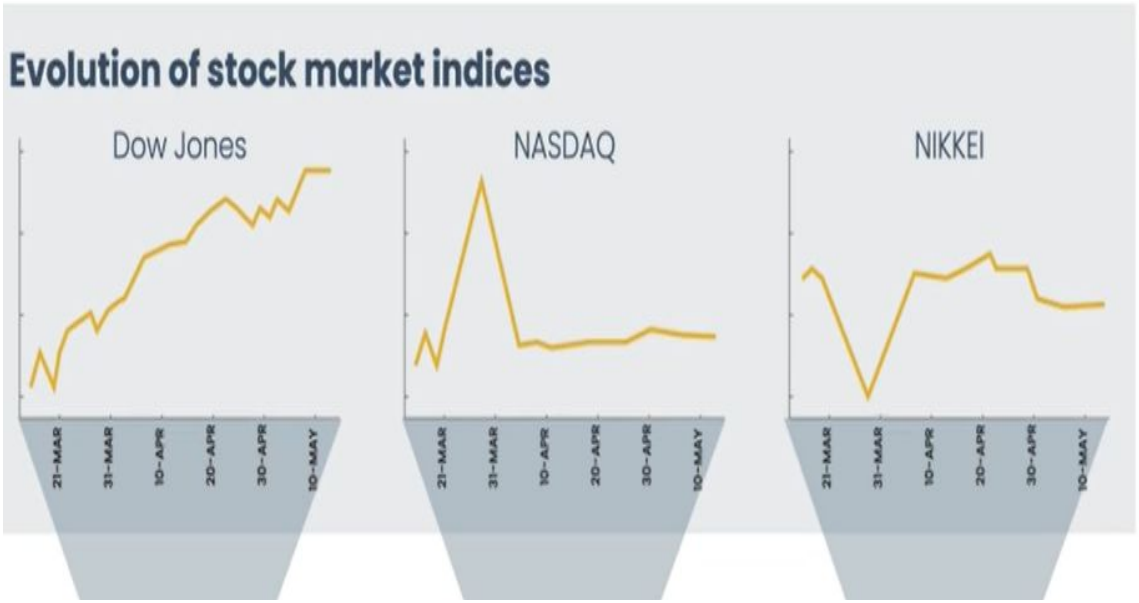


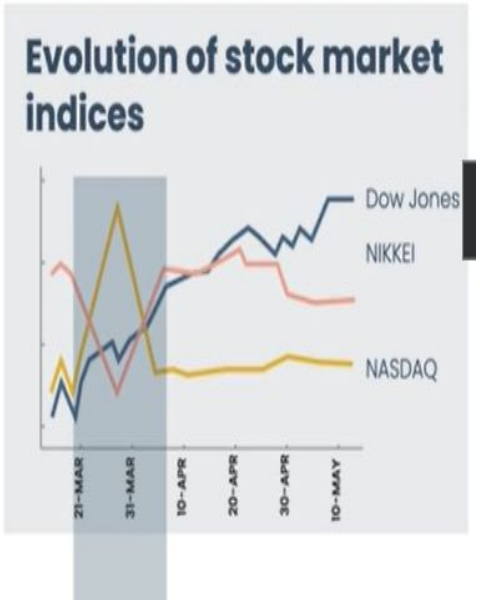


Which One is Better

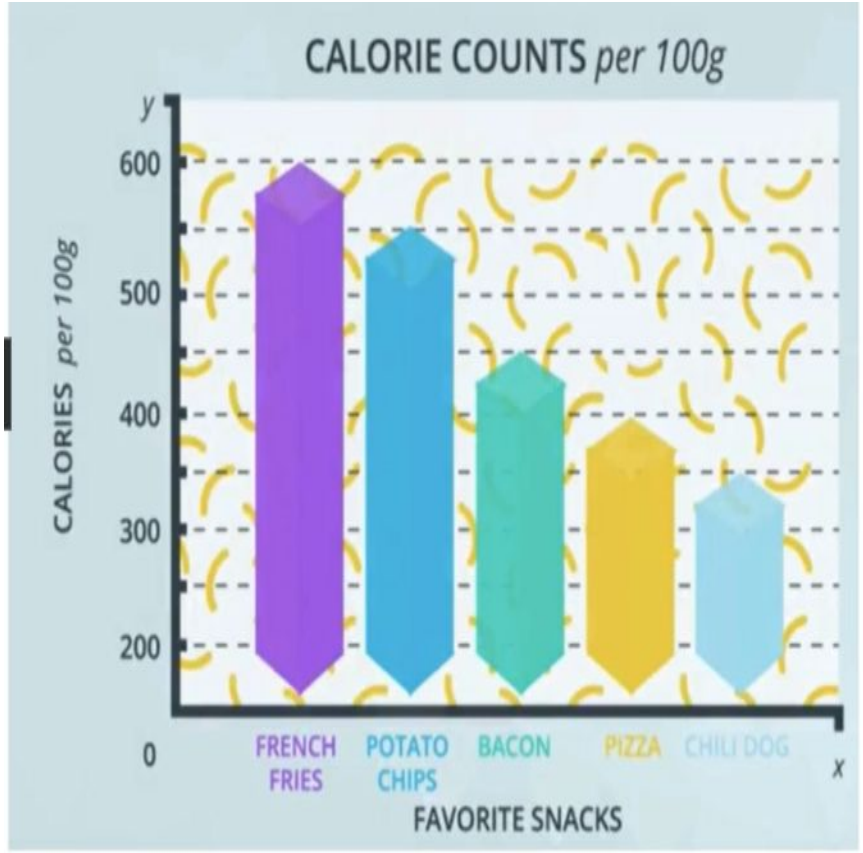
Encourage the eye to compare different pieces of data

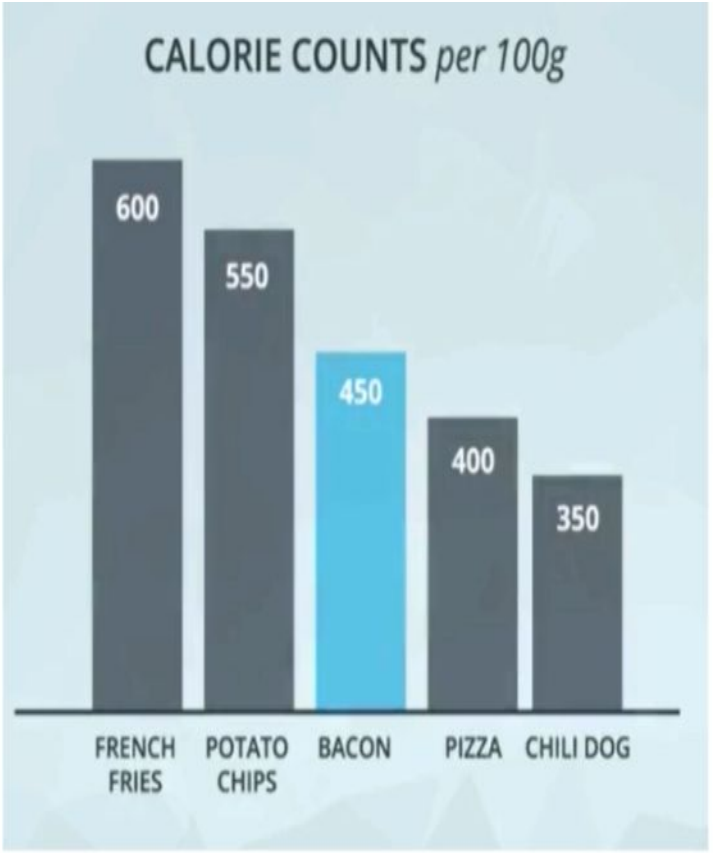
- The goal of a visualization is to help the viewer to think about the data. Your design can help to do that.





Which One is better?





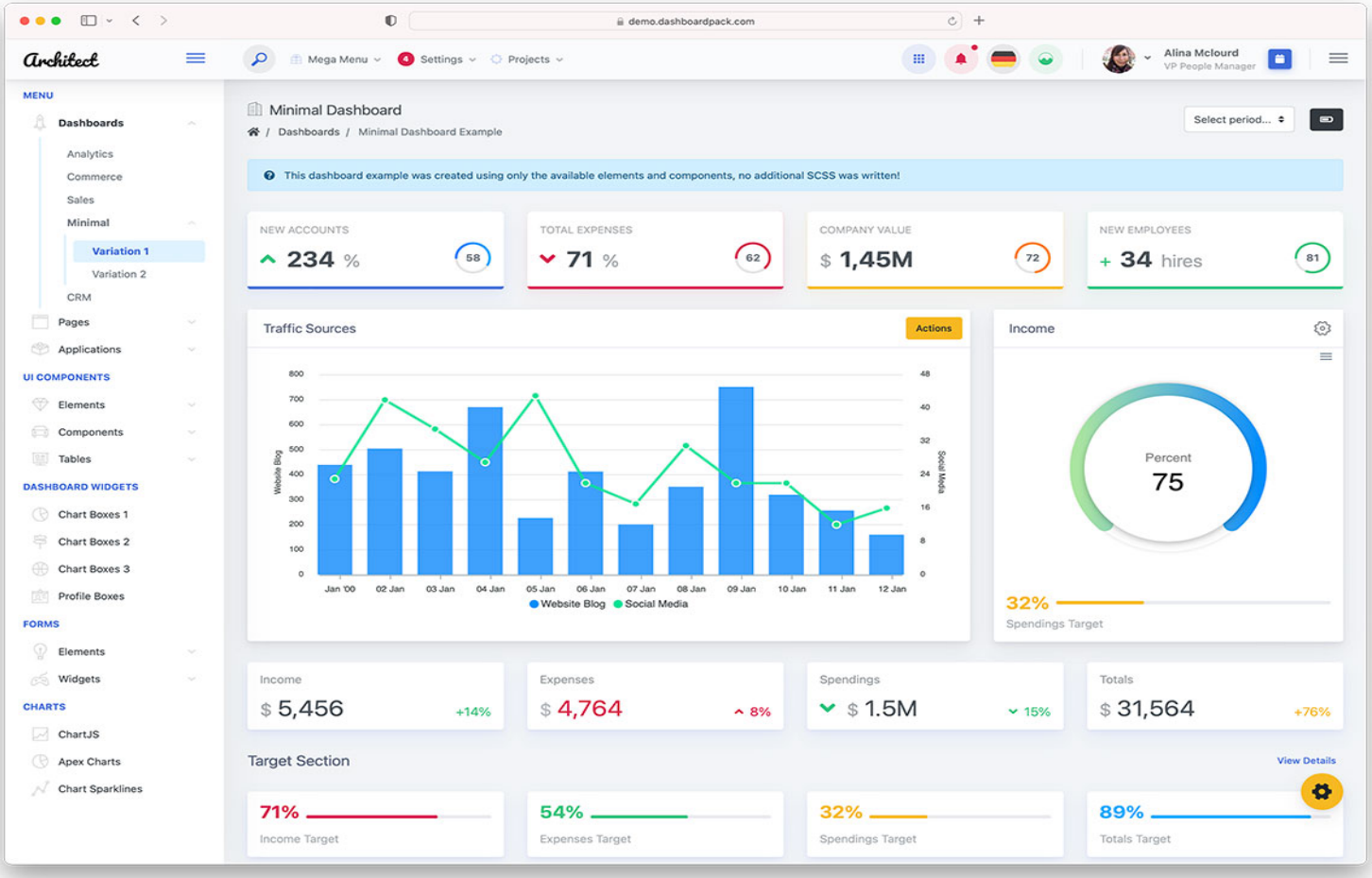
Dashboards

- Data dashboards are a summary of different, but related data sets, presented in a way that makes the related information easier to understand.
- Dashboards are a type of data visualization, and often use common visualization tools such as graphs, charts, and tables.
- Dashboards take data from different sources and aggregate it so non-technical people can more easily read and interpret it.
- With interactive elements, it helps anyone using the dashboard better understand certain points, explore areas of increased interest, and support more questioning to arrive at key insights or make key decisions.

Dashboards

- The main use of a dashboard is to show a comprehensive overview of data from different sources.
- Dashboards are useful for monitoring, measuring, and analyzing relevant data in key areas.
- Used to analyze complex data or subject matter experts to track or present data to non-subject matter experts.
- Dashboards are important because they provide a platform for people to make better, more informed, data-driven decisions.

Dashboards


demo.dashboardpack.com
Architect
Mega Menu
Settings
Projects
Alina Mclourd
VP People Manager
MENU
Dashboards
Analytics
Commerce
Sales
Minimal
Variation 1
Variation 2
CRM
Pages
Applications
UI COMPONENTS
Elements
Components
Tables
DASHBOARD WIDGETS
Chart Boxes 1
Chart Boxes 2
Chart Boxes 3
Profile Boxes
FORMS
Elements
Widgets
CHARTS
ChartJS
Apex Charts
Chart Sparklines
Minimal Dashboard
Dashboards / Minimal Dashboard Example
Select period...
This dashboard example was created using only the available elements and components, no additional SCSS was written!
NEW ACCOUNTS
234 %
58
TOTAL EXPENSES
71 %
62
COMPANY VALUE
$ 1,45M
72
NEW EMPLOYEES
+ 34 hires
81
Traffic Sources
Actions
Website Blog
Social Media
Jan '00
02 Jan
03 Jan
04 Jan
05 Jan
06 Jan
07 Jan
08 Jan
09 Jan
10 Jan
11 Jan
12 Jan
Income
Percent
75
32%
Spendings Target
Income
$ 5,456
+14%
Expenses
$ 4,764
8%
Spendings
$ 1.5M
15%
Totals
$ 31,564
+76%
Target Section
View Details
71%
Income Target
54%
Expenses Target
32%
Spendings Target
89%
Totals Target

Dashboard

